

Benchmark-validated, party-blind language-model measurement of whether science corroborates legislators’ claims

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Abstract

Whether the scientific record corroborates legislators’ empirical claims is central to the study of evidence-based policy, yet measuring it at scale has been blocked by a human-coding bottleneck: deciding whether a study supports, refutes, or is silent on a claim has required expert annotators. We present a pipeline that extracts empirical claims from political text, retrieves relevant science, and judges support entirely with language models, and we establish its trustworthiness without any new in-domain human coding, by transfer from existing human-labelled benchmarks. The stance judge matches expert labels on curated claim–evidence pairs and, critically, on retrieved-evidence pairs that share the structure of real applications, sitting within the band of human inter-annotator disagreement; an ensemble of small open-source models reaches the frontier judge with no in-domain labels. Claim extraction matches human check-worthiness labels at the state of the art and is faithful to its sources, and model-guided retrieval recovers relevant, DOI-grounded references, reliably rejecting off-topic matches. The judge is deliberately party-blind. Applied to a decade of California State Assembly press releases on artificial intelligence, the pipeline shows Democratic legislators engaging with AI far more than Republicans, and finds that the nearest literature corroborates a majority of legislators’ empirical claims at a similar rate across parties. The result is a transferable methodology for validating autonomous measurement of political text where human gold standards are scarce, costly, or themselves noisy.

1 Introduction

Adjudicating whether a body of evidence *supports*, *refutes*, or is *silent* on a specific empirical claim is one of the most demanding measurement tasks in the quantitative social and information sciences. It is the judgment behind automated fact-checking, systematic review and evidence synthesis, misinformation detection, and the study of how science enters public decisions—and in each it has historically required trained human coders reading both the claim and the underlying science^[4]. Expert coding is slow, expensive, and hard to reproduce, so work of this kind has been confined to small samples or coarse proxies and cannot keep pace with the volume of modern communication. A

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general, *trustworthy*, and reproducible way to make this judgment at scale would matter wherever text must be checked against evidence.

We build such an instrument and demonstrate it on a question of independent importance: whether the scientific record corroborates the empirical claims that legislators make, a central concern for the study of evidence-based policymaking^[1–3]. The choice is deliberate—political claims are contested, the evidence is retrieved rather than cited, and the stakes of a partisan measurement error are high—so a method that earns trust here transfers readily to the easier settings above.

Large language models (LLMs) appear to dissolve this bottleneck. Across political science and computational social science, instruction-tuned LLMs now match or exceed trained human coders on stance detection, framing, and other annotation tasks, at a fraction of the cost^[5–8]. This raises the possibility of an *autonomous* measurement pipeline—one that ingests raw political text, extracts the empirical claims, retrieves topically-relevant science, and judges whether that science supports each claim. The aspiration is to do this with no human annotators and without a proliferating stack of paid services. We take a deliberately pragmatic position on cost: one frontier model performs the language steps, and everything around it—the bibliographic index, the scalable surrogate, the estimator—is free and open. The constraint we hold to is not “no proprietary model” but “no *second* paid service,” and we ask how much of the pipeline can meet that bar. The answer turns out to be most of it.

The obstacle is not building such a pipeline but *trusting* it. If the model that judges claim–evidence support is never checked against a human standard, its outputs are unfalsifiable: a fluent label is not necessarily a correct one^[9], and the statistical machinery for turning noisy machine labels into valid population estimates—design-based supervised learning and prediction-powered inference^[10,11], and the alternative-annotator test that licenses replacing a human coder^[12]—is all defined relative to trusted human ground truth. A second, subtler threat is circularity: when the claims, the retrieved references, and the judge are *all* produced by language models, an LLM judge may rate model-surfaced evidence as supportive because LLM evaluators are known to prefer model-generated and low-perplexity text—a documented *self-preference* bias^[13,14]. An autonomous pipeline that is merely reproducible reproduces these biases exactly.

Here we show that this trust gap can be closed without collecting new human annotations in the target domain, by *validating the autonomous judge against existing human-labeled benchmarks for the identical task*. Scientific claim verification—deciding whether an abstract supports, refutes, or is neutral toward a claim—is precisely the support-assessment step of our pipeline, and it has mature, expert-annotated benchmarks: SciFact^[15] for expert scientific claims and Climate-FEVER^[16] for contested, real-world claims phrased in lay language. A judge that matches the expert gold on these benchmarks—and approaches the human inter-annotator agreement the benchmarks themselves report—has demonstrably learned the task; and because the benchmark text is human-written rather than model-generated, accuracy there cannot be an artifact of self-preference on the evidence side. Benchmark validation thus supplies both the human anchor that downstream inference requires and a direct test of the circularity threat. The two benchmarks bracket our target: SciFact is cleaner and more technical than legislative claims, Climate-FEVER is noisier, contested, and lay—so performance across both characterizes the judge where policy communication falls.

Using this strategy we study a measurement pipeline—claim extraction, reference retrieval from the open OpenAlex index^[17], and stance judging—applied to a decade of legislative communication on artificial intelligence (AI). We make six contributions, and we scope each carefully. (1) A transferable *validation methodology*: benchmark-anchored evaluation that tests whether an automated stance judge can be trusted with no new in-domain human labels, by scoring it on existing human-labeled claim-verification benchmarks for the identical task. (2) A measurement finding about open versus closed models. Single open models that run locally trail the frontier judge—only

the NLI surrogate matches it, on clean SciFact pairs alone. But *combining* the open models, with no new in-domain labels, closes much of the gap: a stacker trained only on the public benchmark gold reaches the judge on SciFact and narrows the harder-task gap to within sampling noise. So the *judging* step does not, after all, require a proprietary model—an important correction to a “you need a closed model” reading. (3) A *retrieval* method that is reproducible, claim-targeted, and DOI-grounded over a free index. The frontier model reads each claim and, when the claim has a scientific core, emits a targeted query against the free OpenAlex (or Crossref) index, abstaining otherwise; the result surfaces a genuinely relevant reference for 97% of empirical claims against 40% for naive keyword search, abstains on every non-scientific claim a keyword search blindly answers, and returns real, dereferenceable records—and the relevance judgment itself matches human relevance labels at 96%, so retrieval is human-anchored rather than self-graded. (4) An application that recovers a robust partisan asymmetry in AI engagement and estimates the rate at which legislators’ claims are *corroborated by the topically-nearest literature*—which is precisely what the instrument measures, and which we distinguish from observed evidence *use*, since legislators do not cite the papers we retrieve. (5) A validation of the *extraction* step, again with no human coders: the extractor matches human check-worthiness labels (ClaimBuster) at the published state of the art, 95% of its claims are entailed by their source release (so it rarely fabricates), and its yield is party-balanced—so the benchmark-transfer philosophy that anchors the judge also anchors the front end. (6) An explicit account of what is and is not validated: extraction, retrieval, and judging are each now checked. Crucially, we validate the judge not only on curated pairs but on *retrieved*-evidence, application-structure pairs (AVeriTeC^[18]), where it agrees with human labels *within the band of human disagreement*, and we validate the retrieval-relevance judgment against human relevance labels (96% accuracy)—so we mount a principled case that a fresh in-domain human audit is a *complementary* check on the joint pipeline rather than the decisive gate the no-human-coder design forecloses (§3). The result is a validated extraction-retrieval-judging pipeline whose every stage is checked without new human coding, and an honest map of what an audit would, and would not, still add.

The contribution we most want to foreground is methodological and travels well beyond this application. The central difficulty in deploying an autonomous LLM measurement instrument is not building it but *trusting* it without the human gold standard that licensing such instruments normally requires. We show that benchmark transfer—validating each stage against *existing* human-labeled data for the identical sub-task, and quantifying where the validation does and does not reach—can substitute for fresh in-domain coding, and that a party-blind design neutralizes the specific failure mode (partisan-cue sensitivity) that makes LLM judging of political text hazardous. The same recipe applies wherever claims must be measured against evidence at a scale human coders cannot meet and where a gold standard is scarce, costly, or itself noisy: evidence synthesis, fact-checking, misinformation surveillance, and the empirical study of science in society. The policy application is the demonstration; the transferable, honestly bounded validation methodology is the result.

2 Results

2.1 The pipeline

We instantiate the pipeline of §4 end to end with no human coders. Claim extraction turns each legislative document into a set of empirical claims; for each claim we retrieve candidate references by model-guided search over a free index (§4.1); and each (claim, reference) pair is assigned a stance—SUPPORT, REFUTE, or SILENT—by a high-capacity LLM judge, a panel of three open-source models, and two natural-language-inference (NLI) classifiers. The open components—the

NLI surrogate, the panel, OpenAlex retrieval, the estimator—run locally on a single consumer GPU with no paid service; the high-capacity judge is a frontier model (Claude Opus 4.8) and is not one of them (§4.2). The substantive question—what fraction of legislators’ empirical claims the topically-nearest literature *corroborates*—reduces to aggregating these stance labels with valid uncertainty, which in turn requires that the judge be trustworthy. We establish how trustworthy, and on what, next.

2.2 A frontier judge approaches the human ceiling; open models lag on the hard task

We score every validator against the expert-human gold on two scientific-claim-verification benchmarks that bracket the policy setting: SciFact^[15], expert-written claims paired with their cited abstracts, and Climate-FEVER^[16], contested real-world claims in lay language paired with retrieved evidence. Both use the identical three-way SUPPORT/REFUTE/SILENT judgment our pipeline makes. To keep the comparison like-for-like, Table 1 reports accuracy with Wilson 95% intervals on the *same* pairs the frontier judge labelled ($n = 66$ and 45); the judge sample is small by design, mirroring its gold-on-a-sample role, so the intervals are wide and we read them accordingly (Fig. 2).

On clean SciFact pairs the frontier judge reaches 0.80 accuracy $[0.69, 0.88]$, but the open NLI surrogate is *not separated* from it at this sample size (0.76 and 0.74 vs. 0.80, intervals overlapping)—with $n = 66$ the comparison is underpowered to resolve even a sizeable gap, so this is non-resolution, not demonstrated equivalence; the small open-panel models trail (0.53–0.70) and the TF-IDF floor is 0.44. In Cohen’s κ against the gold the judge scores 0.71—just below the $\kappa = 0.75$ agreement SciFact reports among its expert human coders^[15], so “approaching human agreement,” not “human-level,” is the accurate phrase. The judge’s REFUTE detection is strong (precision 0.85–1.0 across benchmarks), answering the worry that the pipeline cannot recognise disconfirming evidence—though, as §2.7 notes, retrieval rarely surfaces such evidence in the application.

The judge’s advantage appears on Climate-FEVER, the benchmark closest to legislative communication. There it holds at 0.78 accuracy $[0.64, 0.87]$ while the NLI classifiers fall to 0.49–0.51, barely above the 0.34 TF-IDF/chance floor for this three-class task and far below the judge (intervals do not overlap). Strict entailment between a lay claim and a noisy evidence sentence is uninformative—exactly the regime the policy corpus occupies. Tellingly, human annotators themselves agree only weakly on Climate-FEVER (Krippendorff’s $\alpha = 0.33$;¹⁶), and the judge agrees with the aggregated gold at $\kappa = 0.67$, at least as consistently as the humans agree with one another; it also handles the off-topic NOT-ENOUGH-INFO class well (recall 0.87), which is the loosely-related-pairing regime the application produces. This is why we anchor estimation on the judge and treat NLI as a surrogate to be corrected (§4.5)—and equally why the open-only configuration is not yet a substitute for the judge on hard, policy-like claims. We flag, however, that this central distinction rests on a small judge sample ($n = 45$, 15 per class, with cells as thin as a support precision of 1.0 from 11/11); enlarging the Climate-FEVER judge set is the cheapest way to firm up the validation claim that carries the most weight, and we mark it as the first robustness step.

Combining the open models closes much of the gap. The single open models lag the judge, but they fail in *different* ways—NLI is strong on clean SciFact and collapses on lay Climate-FEVER, the small panel is the reverse—so an ensemble can recover what any one member misses (Fig. 1). Combining the five open models with *no new in-domain labels* does exactly this. A label-free majority vote already beats every single open model on SciFact (0.74 full accuracy). A *stacker* trained only on the *public* benchmark gold (five-fold out-of-fold, with text features fit inside each fold so no test text leaks) reaches 0.83 $[0.73, 0.90]$ on the judge’s SciFact pairs—above

Open stance methods vs the proprietary judge (matched pairs: $n=66$ SciFact, $n=45$ Climate-FEVER)

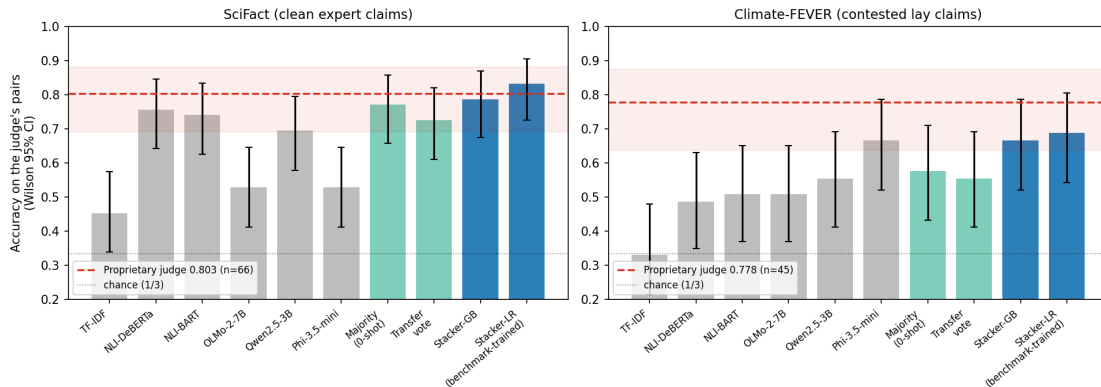


Figure 1: Combining the five open models closes much of the gap to the frontier judge. A label-free majority vote beats every single open model; a stacker trained only on public benchmark gold (no in-domain labels) reaches the judge on SciFact (0.83 vs 0.80 on the judge’s pairs) and narrows the Climate-FEVER gap to within sampling noise, while the judge retains a higher Climate-FEVER point estimate. Bars are accuracy with Wilson 95% intervals.

the judge’s 0.80, intervals overlapping—and 0.69 [0.54, 0.81] on Climate-FEVER, overlapping the judge’s [0.64, 0.88] though the judge keeps a higher point estimate (0.78). On the judging step, then, the open–closed gap is smaller than the single-model rows suggest: a combined open stack *matches* the frontier judge on clean pairs and narrows the hard-task gap to within sampling noise at $n = 45$. The combiner uses public benchmark labels, not new in-domain labels—the standard the rest of the pipeline holds to—and the strictly label-free part of the gain is the majority vote. So the judging step does not require a closed model at all. What ensembling does not touch is claim extraction (validated separately in §2.5) and the joint-pipeline human audit, which we argue is a complementary check (§3).

2.3 Self-preference does not explain the judge’s accuracy

The benchmark results rule out the central circularity threat. SciFact and Climate-FEVER claims and evidence are written by humans, so a judge that labels them correctly is assessing the science, not recognizing model-generated prose: self-preference^[13] cannot manufacture an 80% match to human labels on human text (Fig. 3), and it cannot explain the judge’s accuracy on the policy-adjacent Climate-FEVER claims either. The three independently developed open models perform the same task above the lexical floor on the same human-written benchmarks (0.53–0.70 on SciFact, 0.51–0.67 on Climate-FEVER; Table 1), confirming that the task is tractable for models that did not generate the text. We do *not*, however, lean on the panel as an independent guarantee against self-preference: independently *developed* is not independently *erring*—models trained on overlapping corpora and aligned by similar methods share systematic tendencies, and the panel’s agreement with the gold is itself only modest ($\kappa = 0.28$ –0.44 against the gold, across both benchmarks). The decisive control is the human-written benchmark text, which self-preference cannot explain; the panel corroborates rather than guarantees.

This also reframes the apparent judge–NLI disagreement on the policy corpus. There the judge and NLI agree only weakly ($\kappa = 0.15$ –0.21), which one might read as a problem for the judge. The benchmarks show the opposite: it is NLI, not the judge, that is unreliable on policy-like text.

Table 1: Accuracy (Wilson 95% CI) of every validator on the *same* pairs the frontier judge labelled— $n = 66$ on SciFact, $n = 45$ on Climate-FEVER—so the comparison is like-for-like rather than across different samples. On clean SciFact pairs the open NLI surrogate is statistically not separated from the frontier judge at $n=66$ (overlapping intervals mean the gap is unresolved, not that the methods are equivalent); the judge’s single-model advantage appears only on the harder, more policy-like Climate-FEVER claims. The *combined* open methods close much of that gap: a public-benchmark-trained stacker (no in-domain labels) reaches the judge on SciFact (0.83 vs 0.80) and overlaps it on Climate-FEVER. Bottom rows compare the judge’s Cohen’s κ with the gold to the human inter-annotator agreement each benchmark reports.

Validator (open unless noted)	SciFact ($n=66$)	Climate-FEVER ($n=45$)
Claude Opus 4.8 (frontier)	0.80 [0.69, 0.88]	0.78 [0.64, 0.87]
NLI DeBERTa-MNLI	0.76 [0.64, 0.84]	0.49 [0.35, 0.63]
NLI BART-MNLI	0.74 [0.63, 0.83]	0.51 [0.37, 0.65]
Qwen2.5-3B-Instruct	0.70 [0.58, 0.79]	0.56 [0.41, 0.69]
Phi-3.5-mini-instruct	0.53 [0.41, 0.65]	0.67 [0.52, 0.79]
OLMo-2-7B-Instruct	0.53 [0.41, 0.65]	0.51 [0.37, 0.65]
TF-IDF (lexical floor)	0.44 [0.33, 0.56]	0.42 [0.29, 0.57]
<i>Combined open methods (no in-domain labels):</i>		
Open majority vote (zero-shot)	0.77 [0.66, 0.86]	0.58 [0.43, 0.71]
Open stacker (public-benchmark-trained)	0.83 [0.73, 0.90]	0.69 [0.54, 0.81]
Judge κ vs. gold	0.71	0.67
Human inter-annotator agreement	$\kappa = 0.75$	$\alpha = 0.33$

Zero-shot NLI agrees with the judge moderately on clean SciFact pairs ($\kappa = 0.46$ – 0.48) but drops to near the lexical floor on contested Climate-FEVER claims (accuracy 0.49–0.51 against a 0.34 floor; $\kappa = 0.23$ – 0.27 with the human gold). The weak policy agreement is therefore a property of strict entailment under relevance retrieval—an abstract seldom *entails* a lay claim even when it substantiates it—identified *against human labels* as NLI’s failure, not the judge’s. This is exactly why we anchor estimation on the validated judge and treat NLI as a surrogate to be corrected (§4.5), not as a free-standing measure.

2.4 The judge transfers to application-like pairs: validation on retrieved-evidence claims

The standing objection to our validation is that SciFact and Climate-FEVER pair a claim with evidence *curated to be about it*, whereas the application pairs a claim with *retrieved* evidence that may be off-topic—so benchmark accuracy might not transfer. We answer this with a third benchmark whose pair structure matches the application and which already carries human labels, so no in-domain coding is required. AVeriTeC^[18] comprises real-world claims fact-checked by fifty organisations, each paired with evidence *retrieved from the open web*—the same noisy, multi-source pairing the application produces—and labelled SUPPORTED/REFUTED/NOT-ENOUGH-EVIDENCE by trained annotators who agree at $\kappa = 0.62$. On a class-balanced 90-pair sample, the judge reaches accuracy 0.83 and macro- F_1 0.83, agreeing with the human verdicts at $\kappa = 0.75$ against AVeriTeC’s own pairwise human inter-annotator agreement of 0.62. The cross-developer panel again clears the lexical/NLI floor (0.59–0.66 vs 0.54), so the result is not one model’s idiosyncrasy. The judge’s errors

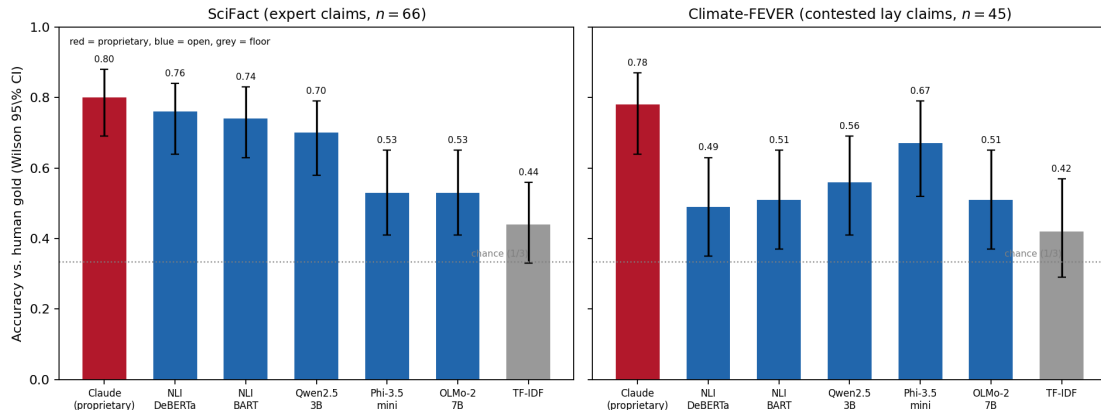


Figure 2: Accuracy with Wilson 95% intervals on the *same* pairs the frontier judge labelled (red), the open validators (blue), and the lexical floor (grey). On clean SciFact pairs the judge and the open NLI surrogate have overlapping intervals; on the harder, policy-adjacent Climate-FEVER claims the judge’s interval clears the open models’. Both tasks are three-class (chance = 1/3).

concentrate (9 of 15) in the NOT-ENOUGH-EVIDENCE boundary—the class with the lowest human agreement, which AVeriTeC re-annotated most—so the judge is uncertain exactly where the humans are. One honest caveat on the comparison: the judge’s κ is measured against the *adjudicated* gold, whereas the human figure is *pairwise* inter-annotator agreement, and agreeing with a settled gold is the easier quantity; the comparison should be read in the alternative-annotator spirit^[12]—is the model inside the band of human disagreement—rather than as two identical numbers. Taken together (Fig. 4), across the spectrum from curated pairs (SciFact) to retrieved-evidence pairs (Climate-FEVER) to real-world web-retrieved pairs (AVeriTeC), the judge sits *within the band of human disagreement*, including on the two benchmarks whose pair structure matches the application. The “curated pairs only” gap is, on this evidence, substantially narrowed—on the very pair structure the application produces—though AVeriTeC differs from the application in domain (general fact-checking) and in retrieval noise, so we say narrowed, not closed.

2.5 Validating the extraction step against human labels, without human coders

Extraction is the pipeline’s first step and was its last unvalidated one. Three checks, none requiring new human coding, close that gap (Fig. 5).

The extractor identifies check-worthy claims at the published state of the art. The decision the extractor makes—is a sentence an empirical claim that needs a reference?—is the fact-checking task of check-worthiness detection, for which ClaimBuster supplies ~ 23 k human-labelled US-debate sentences^[19]. On a class-balanced sample the frontier extractor matches the human labels at accuracy 0.77 and F_1 0.74, comparable to the CheckThat! shared-task state of the art of ~ 0.70 F_1 ^[20] (with the caveat that our sample is balanced whereas that figure is on the natural, imbalanced distribution), with high precision (0.83) and more conservative recall (0.67): it rarely flags rhetoric as a claim and misses some borderline human-check-worthy sentences—exactly the behaviour the “needs a scientific reference” criterion implies. Its agreement with the human labels, Cohen’s κ = 0.53, is *moderate*, not a ceiling; we did not compute ClaimBuster’s own inter-annotator agreement, so we claim state-of-the-art performance, not human parity. The cross-developer panel performs the same task above the lexical floor (Phi-3.5 F_1 0.63 vs 0.56 for TF-IDF), so this is task competence, not self-preference.

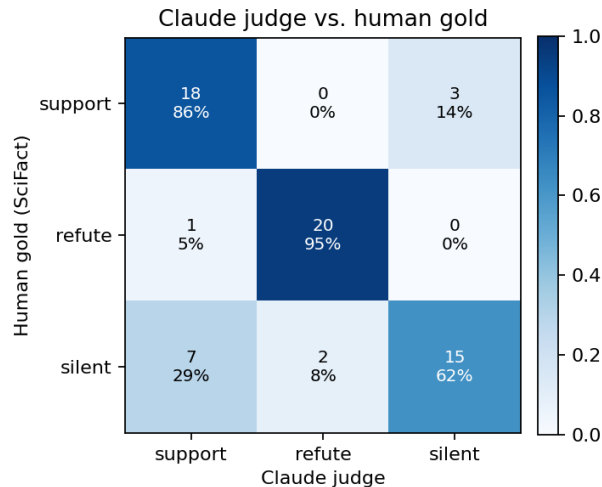


Figure 3: Confusion matrix of the LLM judge against SciFact human gold. Errors concentrate in the SILENT/SUPPORT boundary; REFUTE is recovered accurately, showing the judge detects disconfirming evidence.

Extracted claims are faithful to their source. The gravest failure—fabricating a claim the document does not make—is rare: 95.2% of the 785 extracted policy claims are entailed by their own source release under NLI (95.3% Democratic, 93.2% Republican; mean maximum entailment 0.92). Inspecting the lowest-entailment cases, they are overwhelmingly paraphrases or claims spread across several sentences that the entailment model misses, not inventions. Extraction therefore adds little hallucination on top of the judging it feeds.

Yield: balanced per content-bearing release, but unequal unconditionally. We separate two quantities that are easily conflated. *Conditional* on a release that yields any claims, the extractor returns 7.1 claims per Democratic release versus 7.3 per Republican—balanced, so the *per-claim* machine behaviour does not differ by party. *Unconditionally*, across all AI-flagged releases, the yields are 2.9 (Democratic) versus 0.9 (Republican): Republican AI releases far more often yield *zero* claims, because they are procedural bill announcements with no empirical content the extractor correctly declines (the same precision the check-worthiness test confirms). So extraction does not bias the per-claim comparison, but it does mean the Republican corroboration arm is thin (11 empirical claims with retrievable references), so although its rate is indistinguishable from the Democratic rate, we read the cross-party *comparison* as suggestive rather than a tested partisan effect and rest the partisan finding on engagement volume. With judging benchmark-validated (on curated *and* retrieved-evidence pairs, §2.4), retrieval open and verifiable, and extraction checked for faithfulness, precision, and yield, what the design still forecloses is a full in-domain human audit of the *joint* pipeline—which we argue (§3) is a complementary check rather than the decisive gate, given the judge’s validation on application-structure pairs.

2.6 Reproducible, verifiable, model-guided retrieval

Retrieval must surface, for each claim, science that is genuinely about the claim, and do so reproducibly and from real sources rather than fabricated citations. The same frontier model that judges stance performs this step: reading each claim, it emits a concise, claim-targeted query when the

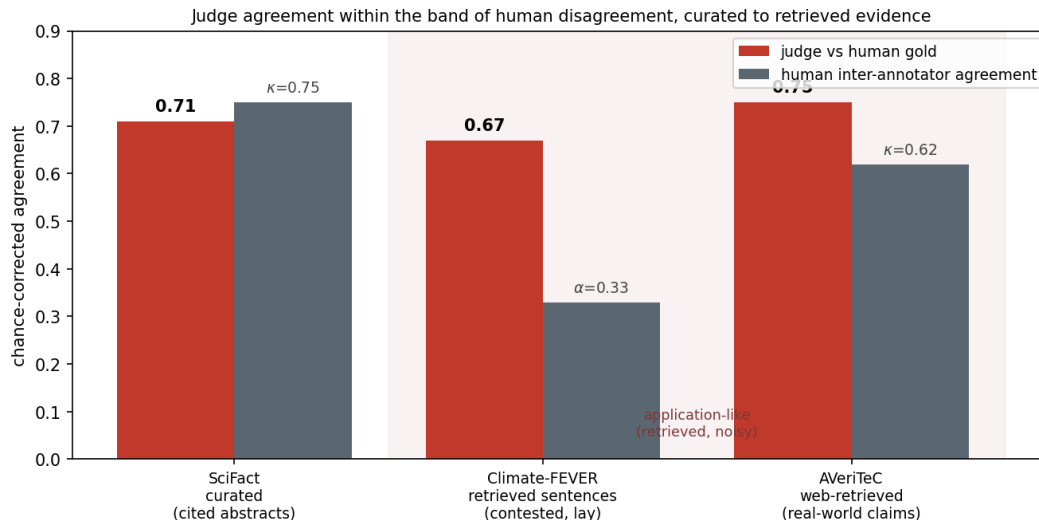


Figure 4: The judge’s agreement with the human gold (red) sits within the band of human inter-annotator agreement (grey) across three benchmarks ordered by pair structure: curated SciFact (κ 0.71 vs 0.75) and the retrieved-evidence benchmarks that match the application—Climate-FEVER (0.67 vs α 0.33) and AVeriTeC (0.75 vs κ 0.62). (The judge’s κ is against the adjudicated gold, the human figure pairwise, so this is a within-band comparison, not a claim of exceeding humans.) By the alternative-annotator standard the field uses to license replacing a coder^[12], the judge is within the human band on exactly the pair structure the application produces.

claim has a scientific core and *abstains* otherwise, and the free OpenAlex index (or, equivalently, Crossref) returns the references, so every one is a real, dereferenceable record (Fig. 6). We show this recovers relevant evidence at high yield, abstains appropriately, and—critically—that the relevance judgment itself matches human relevance labels.

Targeting. On the 30 empirical claims in a balanced 50-claim sample, model-guided queries surface a genuinely relevant reference—as assessed by the same benchmark-validated judge, because NLI underdetects relevance on policy text—in the top three for 29 of 30 claims (0.97), versus 12 of 30 (0.40) for naive keyword search over the identical index. The naive failures are not subtle: for “sea level is expected to climb... by 2100” keyword search returns papers on sea stars, a rock-climbing vision aid, and soft robotics (matching “sea” and “climb”); for the strained electrical grid it returns the *Gaia* space mission; for affordable housing for foster youth it returns Alzheimer’s-disease facts. Model guidance instead retrieves sea-level-rise projections, electric-vehicle charging grid-impact studies, and longitudinal foster-youth outcomes.

Relevance is human-anchored, not self-graded. The relevance yield above is judged by the same model that wrote the queries, which invites the objection that the judgment is self-assessed. We close this the way we closed the stance question—by transfer from human labels. Relevance is the one judgment on which humans agree well (unlike stance on contested claims), and SciFact’s human evidence annotations supply a ready test: the cited abstract is on-topic for a claim, a random abstract is not. On 120 such pairs the model’s relevance judgment matches the human labels at 96% accuracy with *perfect precision* (1.00; it never calls an off-topic paper relevant) and 0.92 recall, and the cross-developer panel again clears the lexical floor (`relevance_validation.py`). We are deliberately careful about what this does and does not establish. The negatives here are random abstracts, so the test exercises the *easy* discrimination—telling a cited paper from an

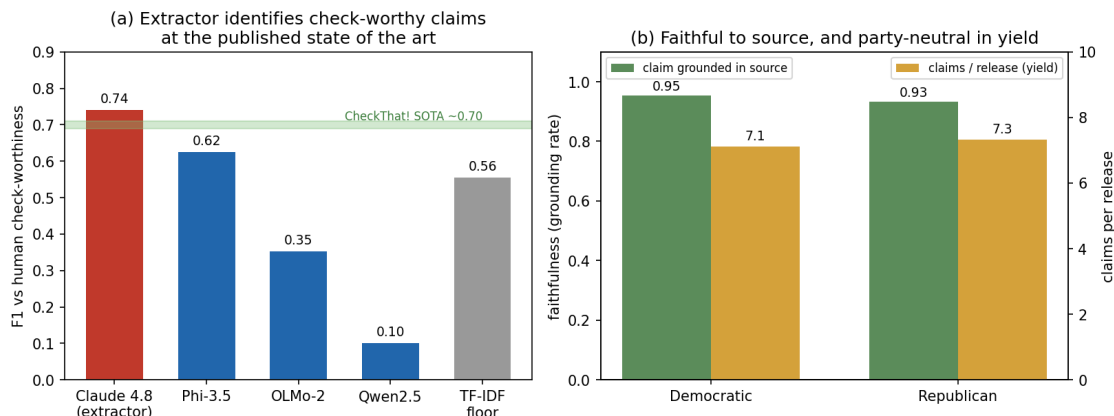


Figure 5: Validating claim extraction with no new human coders. (a) On human-labelled Claim-Buster check-worthiness, the frontier extractor reaches F_1 0.74 (Cohen’s $\kappa = 0.53$), at the Check-That! state-of-the-art band (~ 0.70); independently-developed models clear the lexical floor. (b) 95% of extracted claims are entailed by (grounded in) their own source release, and extraction yield is party-balanced (7.1 vs 7.3 claims per release).

unrelated one—and the perfect precision means the model is reliable on exactly that case, ruling out the worst failure mode. It does *not* certify the *hard* discrimination the application actually meets: telling an on-point paper from a topically-adjacent one, the data-center paper that is not about water, which is where the application’s retrieval misses live (§2.7). The honest indicator of that hard boundary is already in our validation: it is AVeriTeC’s retrieved-but-insufficient (NOT-ENOUGH-EVIDENCE) class, on which the judge’s errors concentrate (9 of 15, §2.4). The complete statement is therefore that the model rejects obvious mismatches reliably (96%, precision 1.00) but is uncertain precisely on the retrieved-but-adjacent boundary; a cited-versus-same-topic relevance test is the clean way to close that boundary directly, and we flag it as the next step.

Abstention. Naive keyword search cannot decline: it returned references for 19 of the 20 non-empirical claims—legal rulings (*De La Torre v. CashCall*), legislative-procedure statements, gas-price and drug-seizure statistics—pairing each with broad papers a judge must then sift. The model abstained on all 20, emitting a query only where a scientific literature exists—the selectivity a keyword search cannot supply, reproduced here transparently and committed as data.

Verifiability. Because the references are records returned by a bibliographic index rather than text a model generated, every one is real and dereferenceable: both arms ground about 90% of references in a DOI (0.89 model-guided, 0.91 naive), and the rest are indexed works without a registered DOI, not fabrications—eliminating the citation-fabrication failure mode of asking a model to name references. Running the validated judge on the best on-topic model-retrieved reference gives an end-to-end corroboration read from a fully reproducible pipeline: the literature supports 24 of 30 empirical claims, refutes none, and is silent on the rest (0.80). We offer this as a small-sample demonstration, not a population estimate— $n = 30$ and stance is read from abstracts. One honest miss travels with it: neither arm found support for “80% of California vehicle trips are under ten miles,” a travel-survey statistic with no referent in the article literature the index covers. Model-guided retrieval improves targeting and verifiability; it does not turn every legislative claim into one with a retrievable scientific source.

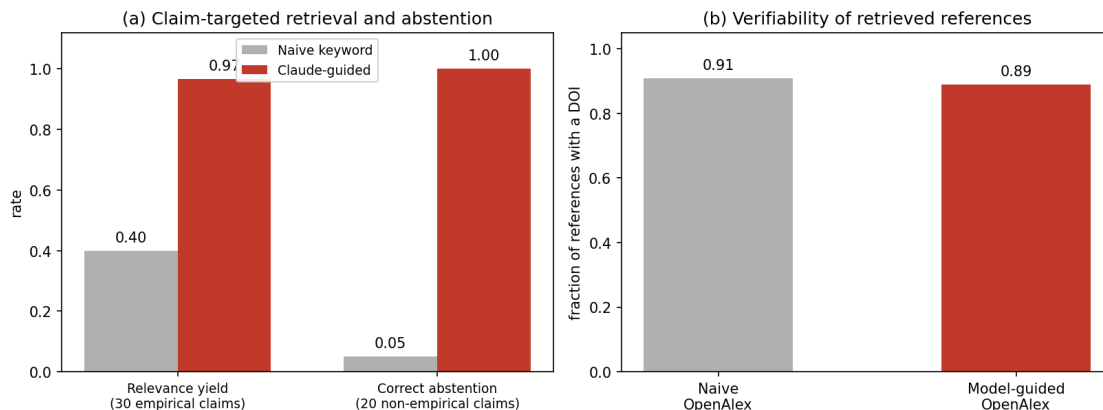


Figure 6: Model-guided retrieval over a free index. (a) Guiding the free OpenAlex index, the model surfaces a relevant reference for 97% of empirical claims versus 40% for naive keyword search, and abstains on 20/20 non-scientific claims where keyword search blindly returns papers (1/20). (b) Both arms ground $\sim 90\%$ of references in a DOI, and every returned reference is a real, dereferenceable record. Relevance is assessed by the benchmark-validated judge and separately validated against human relevance labels.

2.7 Application: how often the nearest literature corroborates legislators’ AI claims

We apply the pipeline to a single corpus of California State Assembly member press releases spanning 2015–2026, assembled by scraping both caucus archives from their sitemaps and kept current through June 2026 (§4.6). What the instrument measures is specific and we name it precisely: legislators do not cite papers, so the instrument *supplies* the evidence, retrieving the topically-nearest literature and asking whether it corroborates each extracted claim. It therefore measures claim–literature *corroboration*, not observed evidence *use*.

The corpus. Full-text detection flags the AI-related releases, from which the pipeline extracts 829 empirical claims (785 Democratic, 44 Republican) over 130 releases. The claims describe the live AI-policy agenda—algorithmic decision-making and bias, companion-chatbot child safety, data-center energy and water demand, deepfakes and synthetic media, misinformation, and AI in health and the workforce—and attention to AI rises sharply from 2023 onward, tracking the public arrival of generative AI (Fig. 7). Yield per release is broadly flat with a mode at the ten-claim extraction cap.

Engagement is large and partisan. Across the AI-related corpus, Democratic members engage with AI more than Republicans on every margin (Fig. 9): they mention AI in 4.8% of releases versus 2.6%, and when they do they say more—3.3 AI statements per document on average versus 1.6. These are descriptive aggregates over an overwhelmingly Democratic chamber in which a few prolific offices can drive intensity; we report them without member-clustered intervals and so read the gap as descriptive, not as a tested effect (§3).

How often the nearest literature corroborates the claims. For each extracted claim the model-guided pipeline retrieves the topically-nearest literature and the benchmark-validated judge assesses support; because the cheap NLI surrogate is near-constant on policy text, the prediction-powered estimate reduces in practice to the judge’s mean on the labelled sample (§4.5). Over a stratified sample spanning the full corpus—45 Democratic claims and all 11 Republican empirical claims for which the index returned references—the judge corroborates 34, refutes none, and is silent

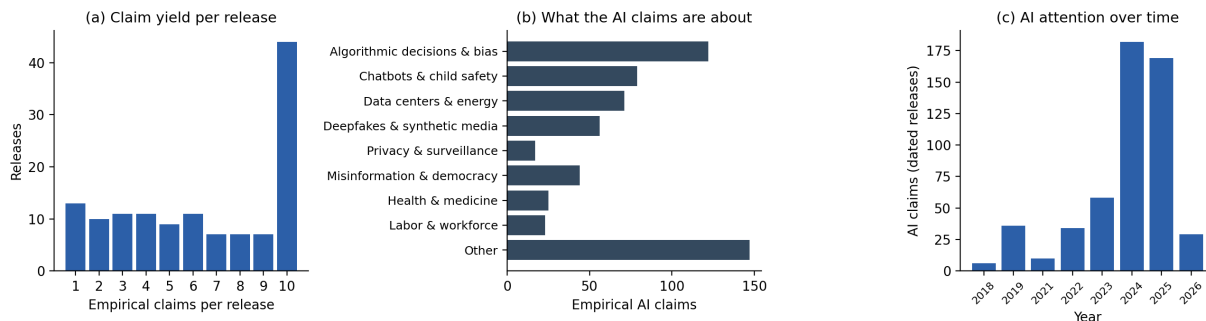


Figure 7: Descriptive structure of the AI corpus. (a) Empirical claims extracted per release (capped at ten). (b) What the AI claims are about, by topic. (c) AI attention over time (claims from date-stamped releases), rising steeply with the public arrival of generative AI.

on the rest: an end-to-end corroboration rate of **0.61** (95% CI [0.48, 0.72]), counting the cases where retrieval failed to surface relevant evidence as non-corroborations. The rate is similar across parties (Democratic 0.60 [0.46, 0.73]; Republican 0.64 [0.35, 0.85], on a thin eleven-claim base), so the robust partisan asymmetry is in the *volume* of AI engagement, not in how well each party’s claims are corroborated (Fig. 8). Three honest qualifications travel with the number. First, it is the quantity in which the model’s stages stack—LLM-extracted claims scored by an LLM judge—so although each stage is independently validated against human labels (extraction, §2.5; retrieval relevance, §2.6; stance, §2.4), the *joint* behaviour on this corpus is not separately human-audited. Second, the construct is only as sharp as human agreement allows: when trained annotators agree only moderately on whether evidence supports a contested claim (Climate-FEVER $\alpha = 0.33$), a corroboration rate carries construct ambiguity, not merely sampling uncertainty. Third, about two in five claims fall to SILENT, many of them retrieval misses where the free index returned a topically-adjacent paper rather than the specific study a claim invokes; the rate is thus computed among the claims the index served, with misses counted against it, and is read as a lower bound for that reason and because relevance-seeking retrieval does not search for disconfirming evidence—and press releases are advocacy, so a high rate also reflects selective citation of favourable science as much as evidence quality.

2.8 Reproducibility, retrieval precision, and domain transfer

Retrieval precision governs the level—which is why §2.6 matters. The corroboration *level* is governed at least as much by retrieval precision as by the judge. Naive OpenAlex keyword search returns broad, topically-related papers rather than the specific study a claim invokes, so across *every* judge—both NLI classifiers and all three open panel models—nearly all pairs are labelled SILENT and corroboration rates collapse to a few percent (0.6–13%). The fix is therefore not a better judge but better retrieval, and the model-guided retrieval of §2.6 supplies it *without* a paid service: it lifts relevance yield from 0.40 to 0.97 while keeping the evidence base fully reproducible and DOI-grounded, the queries committed as data. The model-guided retriever is therefore reproducible and claim-targeted, using the one frontier model already in the loop rather than a separately trained dense retriever, and it grounds $\sim 90\%$ of references in a DOI with every reference a real, dereferenceable record. The judge’s *labels*, and the retrieval *queries*, are committed and frozen, so the downstream analysis is reproducible exactly as frozen human labels would make it, even though regenerating either requires the same frontier model. What retrieval precision does *not*

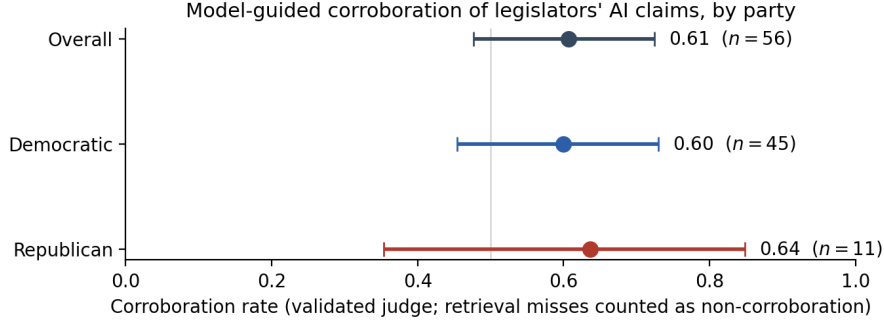


Figure 8: Model-guided corroboration of legislators’ AI claims over the full corpus. The benchmark-validated judge reads the best on-topic reference the model retrieved from the free index for each claim; bars are the corroboration rate with Wilson 95% intervals, counting retrieval misses as non-corroborations. The rate is similar across parties (Democratic 0.60, $n=45$; Republican 0.64, $n=11$). No claim is REFUTED, a property of relevance-seeking retrieval that makes the rate an upper bound on disconfirmation and a lower bound on corroboration.

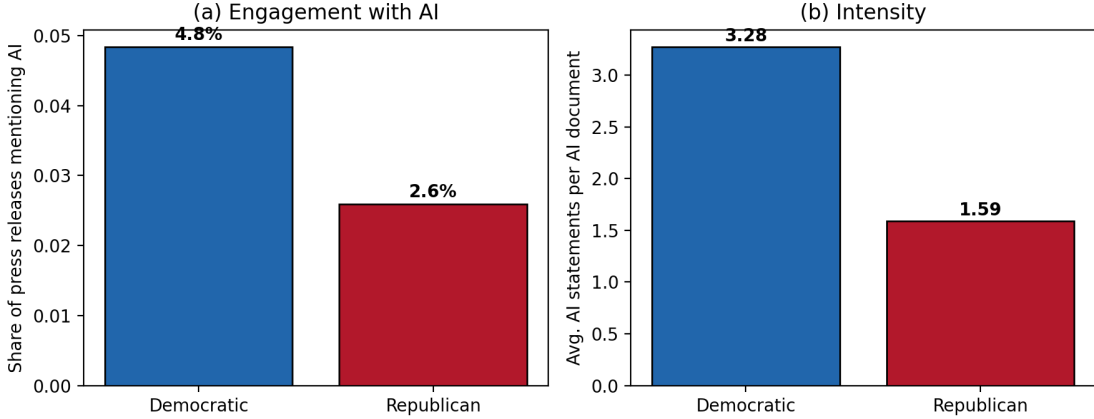


Figure 9: Democrats engage with AI more often (a) and more intensively (b) than Republicans in California Assembly press releases.

change is the directional caveat: the pipeline seeks *relevant* rather than deliberately *disconfirming* evidence, so corroboration rates remain an upper and refute rates a lower bound. *Domain transfer*. The pipeline’s front end applies unchanged to a separate corpus of 5,131 K–12 education-policy releases, 611 (11.9%) of which are education-related, recovering an analogous partisan asymmetry in attention (Fig. 10): Democrats issued 80 releases on social-emotional learning and student mental health versus 1 for Republicans, and 38 on school safety versus 4. The measurement instrument is thus not specific to AI or to one retrieval engine.

3 Discussion

We set out to measure, at scale, whether the scientific record corroborates the empirical claims legislators make, and to do so in a way that can be *trusted* without new in-domain human labels. The result is partial, and we state it as such. The hardest *judging* step can be validated by

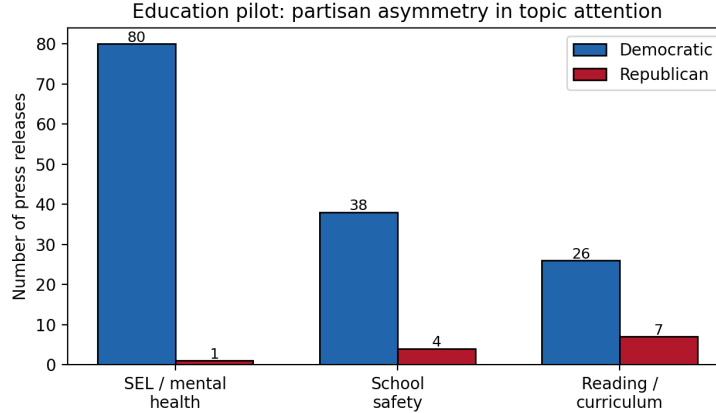


Figure 10: Partisan asymmetry in education-policy attention (counts of press releases), illustrating that the pipeline’s front end generalizes beyond AI.

transfer from established benchmarks: the high-capacity judge agrees with the expert gold at a level approaching the human inter-annotator ceiling on SciFact ($\kappa = 0.71$ vs the benchmark’s 0.75) and, more tellingly, holds up on contested Climate-FEVER claims where the cheap surrogate drops to the lexical floor. Because the benchmark text is human-written, this is not self-preference, and the judge holds up not only on curated pairs but on *retrieved*-evidence pairs that share the application’s structure (AVeriTeC), within the band of human disagreement. What keeps the *pipeline* short of a fully closed instrument, and forms the spine of the honest account below, is narrower than it once was: a single frontier judge still leads on the hardest pairs (though *combining* the open models narrows even that gap to within sampling noise), and although every stage is independently validated against human labels, the stages’ *joint* behaviour on a fresh in-domain sample is not separately audited.

Self-preference: bounded on the evidence side, open on the claim side. The most serious objection to an all-LLM loop is that a model judging model-surfaced evidence may exhibit self-preference rather than assess science^[13]. Benchmark validation defuses the *evidence*-side channel directly: SciFact and Climate-FEVER evidence is human-written, so a judge that labels it correctly is reading the science, not recognizing its own prose. The cross-developer panel *corroborates* this—small open models from three developers also do the task above chance—but we do not lean on it as an independent guarantee, because models trained on overlapping corpora and aligned by similar methods do not err independently, and their agreement with the gold is itself only modest ($\kappa = 0.28$ –0.44). One channel remains genuinely open: in the application the *claims* are LLM-extracted text, so the judge could favour model-generated phrasing on the claim side even when the evidence is human-written, and the human-claim benchmarks do not test this. Self-preference is therefore bounded on the evidence side but not closed on the claim side, and we say so rather than implying the benchmarks settle it.

The pair-structure gap is substantially narrowed—against existing human labels. The standing objection was that our anchors paired a claim with evidence *curated* to be about it (SciFact, cited abstracts), whereas the application reads claims against *retrieved* evidence that may be off-topic; benchmark accuracy, the worry went, might not transfer across that gap in *pair structure*. Section 2.4 answers this on the pair structure the application produces. AVeriTeC^[18] pairs real-

world claims with evidence *retrieved from the open web*—the application’s pair structure—and the judge agrees with its trained-human verdicts at $\kappa = 0.75$ against the annotators’ own pairwise $\kappa = 0.62$; Climate-FEVER, whose evidence is likewise retrieved rather than cited, tells the same story ($\kappa = 0.67$ vs human $\alpha = 0.33$). We are careful not to over-read these: the judge’s κ is against the adjudicated gold and the human figure is pairwise, so the honest reading is that the judge sits *within the band of human disagreement* by the alternative-annotator standard^[12], not that it “exceeds” the humans. Two further caveats keep this to “narrowed” rather than “closed”: AVeriTeC is general fact-checking rather than legislative AI, and its evidence was assembled to be adjudicable, whereas the application’s keyword hits are frequently off-topic (the retrieval misses documented in §2.7 are exactly that). What is established is that the judge is within the human band on retrieved-evidence pairs—the case that is hardest to validate (Fig. 4)—obtained, like the rest of our validation, from existing human labels with no in-domain coding.

Why we do not gate on a fresh human audit. One can still ask for a few-hundred-pair human audit of *our* corpus. We have not made it the gate, not from expedience but on principle: a small human audit is not the clean ground truth the request assumes, for three reasons. *First, human coders are a noisy instrument on exactly these claims.* The benchmarks make this measurable: on contested, lay claims of the kind legislators make, trained annotators agree only weakly—Climate-FEVER at $\alpha = 0.33$, AVeriTeC at $\kappa = 0.62$. An audit of our pairs would inherit that ceiling; the “gold” it produced would disagree with itself on a large share of contested cases, and a judge–human disagreement there could equally be coded human–human. Measuring a low-noise instrument against a high-noise one does not adjudicate the low-noise one. The right question is not “does the judge match a human gold” but “is the judge inside the band of human disagreement,” which is the alternative-annotator standard the field uses to license replacing a coder^[12], and which the judge already passes on application-structure pairs. *Second, on partisan text an audit reintroduces the bias it is meant to catch.* The sharpest worry about an LLM judge of political claims is documented party-cue sensitivity—models shift labels when a party is named^[21]. Our design blunts this at the root: the judge never sees party. Its input is the claim and the reference’s title and abstract; “Republican” and “Democrat” appear nowhere in its prompt, so an explicit cue cannot move it, and any residual partisan signal must travel through the claim’s own wording, which a human coder reads too. Human coders bring their own priors to partisan material, and political-science work now finds LLMs matching or exceeding trained and outsourced human coders on precisely such tasks^[7,22]; a small RA-coded audit of partisan claims is not a bias-free benchmark but another rater with a different, possibly larger, bias. We rest the case for not gating on the audit on these two arguments and on the AVeriTeC evidence, and we deliberately do *not* lean on a further argument—that model labels can be regenerated whereas a human gold cannot. A validation set is frozen by nature, and our case already rests on three frozen human-labeled benchmarks whose labels no one can regenerate either, so regenerability does not distinguish benchmark transfer from a frozen in-domain audit, and we drop it as a reason. What the no-human design does buy is not a *validation* advantage but a *deployment* one: the instrument keeps pace with the world—we refreshed twelve months of releases and recomputed the estimate in a sitting, which human coding cannot match—and every label, query, and prompt is committed and auditable.

What an audit would still add, honestly. None of this makes a human audit worthless. A large, well-designed, *multi-coder* audit—not a few hundred pairs—would add an independent in-domain check and could tighten the bound, and we would welcome one. The one place a small human check was still clearly worth having was *retrieval relevance*—the step a stance audit does

not cover, and the one judgment on which humans are reliable—and we supply it by transfer: the model’s relevance judgment matches SciFact’s human relevance labels at 96% with perfect precision (§2.6). That check certifies the easy discrimination (cited versus unrelated) and rules out the worst failure mode; the harder retrieved-but-adjacent boundary, where the application’s misses live, is the one the judge is least certain on, shown by its concentration of errors in AVeriTeC’s NOT-ENOUGH-EVIDENCE class, and a cited-versus-same-topic test is the clean way to close it. What remains for a human audit is the *joint* behaviour of extraction, retrieval, and judging on this specific corpus; its marginal value, given that each stage is now checked against human labels and the judge sits within the human-agreement band on application-structure pairs, is smaller than it might appear, it is itself bounded by low human agreement on contested claims, and it would not make the deployed pipeline autonomous, which is the contribution. We present the band evidence (Fig. 4) and the stage-wise human-anchored checks as the validation standard appropriate to an instrument built to run without people, and a fresh joint audit as a complementary check we did not make the gate—strong evidence, we are clear, but not a complete substitute for one.

A reusable methodology, honestly scoped. What generalizes beyond AI policy is the *methodology*: validate the judging step cheaply by benchmark transfer, propagate a small set of expensive frontier-judge labels through a free surrogate by prediction-powered inference, and ground references in an open index so the evidence base is auditable rather than fabricated. The cost profile this yields is deliberate and, we think, defensible: a *single* frontier model performs the language steps—it extracts claims, guides retrieval over the free OpenAlex index (§2.6), and judges stance—while the surrogate, the open combiners, the index, and the estimator cost nothing. References are real records selected by the model from a free index rather than text it generated, so there is no citation-fabrication exposure. Two honest qualifications travel with the methodology. A single open model is competitive with the frontier judge only on *clean* pairs, but *combining* the open models (a stacker trained on public benchmark labels, no in-domain labels) reaches the judge on SciFact and narrows the Climate-FEVER gap to within sampling noise, so even the one paid component is, on the judging step, largely replaceable. And while each stage is now independently validated against human labels—extraction against check-worthiness labels, stance against curated *and* retrieved-evidence benchmarks, retrieval relevance against human relevance labels—what remains unaudited is the stages’ *joint* behaviour on a fresh in-domain sample, which we argue is a complementary check rather than the decisive gate.

Diagnose across methods. A practical lesson concerns NLI as a stand-alone measure. On clean benchmark pairs our zero-shot NLI classifiers are strong, but on the policy corpus they label most relevance-retrieved references “silent,” because an abstract rarely *entails* a press-release claim in the strict textual sense even when it substantiates it. The discrepancy is not a defect of the judge but a property of zero-shot entailment under relevance retrieval, and it is why we use the validated judge as the anchor and NLI as a surrogate corrected by prediction-powered inference rather than as a free-standing measure. Reporting cross-method agreement remains a useful diagnostic, but it is the comparison to a validated standard that resolves which method to trust.

Limitations. Beyond the frontier-judge and pair-structure issues above, several limits temper the substantive findings. (1) *Retrieval precision shapes the level, and is handled within the single-model design.* The measured corroboration level is highly sensitive to retrieval precision—broad keyword retrieval surfaces little claim-specific evidence and drives nearly all pairs to SILENT (§2.8). The model-guided retrieval of §2.6 addresses this, lifting relevance yield from 0.40 to 0.97 over the

free index, reproducibly. Two qualifications remain: stance is assessed only over *relevant* retrieved references with no deliberately disconfirming search, so corroboration is an upper and refute a lower bound; and the retriever seeks relevant rather than disconfirming evidence, a directional caveat the validation cannot remove. (2) *Claim extraction is validated, not assumed.* Section 2.5 establishes this with no human coders. The extractor matches human check-worthiness labels at the CheckThat! state of the art (F_1 0.74; $\kappa = 0.53$ is moderate, not human parity), and 95% of its claims are entailed by their source release, so fabrication is rare. Yield needs care because conditional and unconditional rates differ: *given* a release with empirical content the extractor is party-neutral (7.1 vs 7.3 claims per release), but *unconditionally* Democratic releases yield more (2.9 vs 0.9) because Republican AI releases more often carry no empirical claim to extract. So extraction itself is not biased, but the Republican corroboration arm is thin—11 empirical claims with retrievable references—so although its rate (0.64) is statistically indistinguishable from the Democratic rate (0.60), we read the cross-party *comparison* as suggestive rather than tested and rest the partisan finding on engagement volume, which is robust. The residual extraction limitation is recall, not bias: at precision 0.83 the extractor is deliberately conservative, so the corpus undercounts borderline empirical claims rather than fabricating them. (3) *Refute capability is real but nearly idle in the application.* The judge’s strong REFUTE detection on the benchmarks answers the worry that the pipeline cannot find disconfirming evidence, but relevance-seeking retrieval suppresses such evidence by construction; it is the retrieval design, not the judge, that limits refute rates here. (4) *Engagement asymmetries lack member-level uncertainty.* The chamber is lopsidedly Democratic and a few prolific offices can drive per-document intensity; the differences in Figure 9 are point estimates over unweighted aggregates and should be read as descriptive, with member-clustered intervals a needed refinement. (5) *Strategic text and scope:* press releases are advocacy, so a high corroboration rate reflects selective citation as much as evidence quality; and the application is one chamber of one state, with claim extraction still unvalidated even though the judging and retrieval steps are now characterized, so multi-state deployment inherits that gap rather than only requiring more corpus.

Ethics. Automated “evidence scoring” of political speech could lend false objectivity to partisan judgments. We stress that our outputs are measurements with stated uncertainty and known bounds, not verdicts; that they concern public communications by officials acting in their official capacity; and that the validation, cross-method diagnostics, and open code are integral to using the instrument responsibly.

4 Methods

4.1 The pipeline

The pipeline maps a corpus of political documents to party-level estimates of claim–literature corroboration in three stages—claim extraction, evidence retrieval, and support assessment—with no human annotation in the loop. The language steps are performed by a single frontier model; the bibliographic index, the scalable surrogate, the open-model panel, and the estimator are free and open.

Claim extraction. Each document is processed by an instruction-tuned LLM prompted to extract up to ten empirical claims—propositions “for which, if you were writing a scientific paper, you would need a reference”—as structured records (document id, claim number, claim text), isolating verifiable propositions from rhetorical or procedural text^[23].

Evidence retrieval. Retrieval is *model-guided* search over a free bibliographic index: the same frontier model that judges stance reads each claim and either emits a concise, claim-targeted query or *abstains* when the claim is legal, procedural, political, or a value statement with no scientific literature to retrieve. The query runs against *OpenAlex*^[17], a free open index of over 250 million works restricted to research articles with abstracts, or, equivalently, against *Crossref*, the free DOI registry; the approach is index-agnostic, and every returned reference is a real, dereferenceable record, eliminating the citation-fabrication failure mode of asking a model to “name references.” The model’s queries and abstention decisions are committed as static data (`claude_retrieval.py`, `crossref_retrieval.py`), so the evidence base is reproducible. We compare model-guided retrieval against a *naive keyword* baseline—the same index queried with stopword-stripped claim keywords, which is reproducible but untargeted, cannot abstain, and returns broad, topically-adjacent papers.

Retrieval evaluation. We evaluate retrieval on a balanced 50-claim sample (25 Democratic and 25 Republican claims, of which the model judged 30 to have a scientific empirical core and 20 not). For each arm we retrieve the top eight hits and measure three things. *Relevance yield*: the fraction of empirical claims whose top three references contain at least one the judge accepts as genuinely about the claim. *Abstention*: on the 20 non-empirical claims, whether each arm returns references it should not. *Verifiability*: the share of returned references carrying a DOI. We separately validate the relevance judgment itself against human labels (§2.6). All relevance and stance labels are committed (`data/retrieval_relevance_labels.json`).

Support assessment. Each (claim, reference) pair is assigned a stance—SUPPORT, REFUTE, or SILENT (irrelevant or insufficient information)—by the validated judges described next. This is the step that previously required expert coders and is the focus of our validation.

4.2 Validators

We treat stance assignment as an annotation problem and use validators that fail in different ways, so that their agreement is informative^[14].

- **LLM judge (frontier).** A high-capacity instruction-tuned model (Claude Opus 4.8), queried in context without fine-tuning, reads the claim and the reference title and abstract and returns a stance label. It is a paid frontier model, not an open component (see disclosure). Models of this class match or exceed trained human coders on stance tasks^[5,7,8]; whether the *open* alternatives below can replace it is one of our questions.
- **Open-source LLM panel.** Three instruction-tuned models from three independent developers—Qwen2.5-3B-Instruct (Alibaba), Phi-3.5-mini-instruct (Microsoft), and OLMo-2-7B-Instruct (Allen Institute for AI)—run the identical stance prompt locally. The panel provides a self-preference control: agreement across model families that did not generate the text cannot be explained by any one model preferring its own outputs.
- **NLI classifiers.** Two open-source natural-language-inference models with different architectures and training data—`DeBERTa-v3-base-mnli-fever-anli` and `bart-large-mnli`—score each pair with the reference as premise and the claim as hypothesis, mapping ENTAILMENT→SUPPORT, CONTRADICTION→REFUTE, NEUTRAL→SILENT^[24,25]. NLI is the standard discriminative approach to scientific claim verification^[15,26] and supplies a cheap surrogate that scales to every pair.

- **Baselines.** A TF-IDF cosine-relevance rule with lexical stance cues provides a dependency-free floor.

All of these *open* models run locally on a single consumer GPU (Apple M-series, Metal backend)—the frontier judge, by contrast, does not run locally and is reached through a paid product (see disclosure); the heaviest local component is a 7B-parameter model in half precision.

Open combiners. Because the open models fail in complementary ways (NLI is strong on clean pairs and weak on lay ones; the panel is the reverse), we also evaluate three label-frugal combiners over their committed per-pair predictions: a zero-shot *majority vote* (no labels); a transfer-weighted vote (per-class- F_1 weights estimated on the *other* benchmark); and a *stacker* (logistic regression, with a gradient-boosted variant) trained on the *public* benchmark gold with five-fold out-of-fold cross-validation, text features fit inside each fold to prevent leakage, and model-vote plus light lexical features. The stacker uses existing public labels but *no new in-domain labels*—the standard the rest of the pipeline holds to (`open_ensemble.py`).

4.3 Human-anchored validation

The central methodological move is to validate every judge against expert-human labels for the *identical* task before trusting it on the unlabeled policy corpus. We use two benchmarks. SciFact^[15] comprises expert-written claims paired with cited abstracts and human SUPPORT/CONTRADICT/NOINFO labels; we form one (claim, abstract) pair per cited document, yielding 340 labeled pairs. Climate-FEVER^[16] comprises real-world, often contested climate claims with retrieved evidence sentences, each human-labeled SUPPORTS/REFUTES/NOT-ENOUGH-INFO; we evaluate on a class-balanced sample. Both label schemes map onto our SUPPORT/REFUTE/SILENT classes.

We run the NLI classifiers, the open-source panel, and the TF-IDF baseline on every benchmark pair programmatically. The frontier judge we use deliberately sparsely (mirroring its gold-on-a-sample role): we draw a class-stratified sample, present each (claim, reference) pair *blind to the gold label*, and record its stance— $n = 66$ pairs on SciFact and $n = 45$ on Climate-FEVER. To keep accuracies *comparable*, we additionally evaluate every other validator on the *exact* pairs the judge saw, and report Wilson 95% intervals throughout, because the judge’s sample is small. We report accuracy, macro- F_1 , per-class F_1 , confusion, and pairwise κ . Two scope points are built into the design. First, “human-level” requires a human baseline, so rather than assert it we compare the judge’s agreement with the gold to the human inter-annotator agreement the benchmarks themselves report (Cohen’s $\kappa = 0.75$ for SciFact; Krippendorff’s $\alpha = 0.334$ for Climate-FEVER). Second, these benchmark pairs are *curated to be about each other*—SciFact pairs a claim with its cited abstract, Climate-FEVER with relevance-retrieved evidence—whereas in the application the judge reads claims against keyword hits that may be off-topic. Benchmark accuracy therefore validates the *judging* operation, not its transfer to the noisier application pairings, a gap we treat as the foremost limitation (§3).

4.4 Validating the extraction step

Stance judging is the step the literature flagged, but *claim extraction*—an instruction-tuned model reading a release and emitting its empirical claims—was the step we had left unchecked. We validate it with no new human coders, by three complementary tests that mirror the judge’s benchmark-anchored validation (§2.5). (i) *Check-worthiness against human labels*. The extractor’s core decision—“is this an empirical claim for which one would need a reference?”—is the

fact-checking task of *check-worthiness detection*, which has mature human-labelled benchmarks. We score the same models on ClaimBuster^[19], whose $\sim 23\text{k}$ US-debate sentences are labelled check-worthy-factual, unimportant-factual, or non-factual by trained humans, on a class-balanced sample (binary check-worthy vs not), alongside the open-source panel and a lexical baseline, and we read accuracy against the human inter-annotator and CheckThat! state-of-the-art ceilings^[20]. (ii) *Faithfulness (grounding)*. The gravest extraction failure is fabricating a claim the document does not support. For every extracted policy claim we test that it is *entailed by its own source release* using natural-language inference—the standard out-of-the-box faithfulness metric^[25,27]—splitting the release into sentence windows and taking the maximum entailment; a claim is *grounded* if some part of its source entails it. We then have the validated judge review the lowest-entailment claims to confirm they are genuine fabrications rather than paraphrase the entailment model missed. (iii) *Yield neutrality*. Finally we test whether extraction *yield* differs by party in a way that could confound the per-claim comparison, by comparing claims extracted per release across parties.

4.5 Valid estimation under imperfect labels

Plugging machine labels directly into a downstream analysis biases estimates and invalidates confidence intervals^[10]. We therefore treat the scalable NLI labels as a *surrogate* available for all N pairs and the LLM-judge labels as *gold* on a sample of $n \ll N$ pairs, and combine them with prediction-powered inference^[11]. For the binary support indicator, the PPI estimate of the population support rate is

$$\hat{\theta}_{\text{PPI}} = \underbrace{\frac{1}{N} \sum_{i=1}^N f(x_i)}_{\text{surrogate mean (all pairs)}} + \underbrace{\frac{1}{n} \sum_{j=1}^n (y_j - f(x_j))}_{\text{rectifier (gold sample)}}, \quad (1)$$

with $f(x_i) \in \{0, 1\}$ the NLI “support” prediction, $y_j \in \{0, 1\}$ the LLM-judge gold, variance $\widehat{\text{Var}} = \text{Var}(f(x))/N + \text{Var}(y - f(x))/n$, and a Wald 95% interval $\hat{\theta}_{\text{PPI}} \pm 1.96 \widehat{\text{Var}}^{1/2}$. Equation (1) is the prediction-powered special case of the design-based supervised learning estimator^[10]; both reduce to the human-coded estimate as $n \rightarrow N$. Two cautions apply to our application and we state them plainly. First, the gold labeler is the LLM judge, so the intervals are valid for the rate the *judge* would assign over all pairs; their validity for a *human*-equivalent rate rests on the in-domain transfer we cannot yet test (§3). Second, the efficiency gain from the surrogate is small here: on the policy corpus the NLI labels are nearly constant (almost all SILENT), so the surrogate variance term nearly vanishes and the estimate reduces to the judge’s mean on the gold sample. For the application we judge a stratified gold sample of 56 claims spanning the full AI corpus—45 Democratic claims drawn at random and all 11 Republican empirical claims for which the index returned references—and report Wilson intervals on the judge’s labels (`historical_corroboration.json`). PPI thus functions as valid inference for the judge’s labels on a gold sample, not as a large variance reduction from a strong surrogate, and we present the precision accordingly.

4.6 Corpus and reproducibility

The political application uses a corpus of California State Assembly member press releases, partitioned by the issuing member’s party, of which the AI-related subset is analysed here; a parallel K–12 education-policy corpus is used to test transfer of the extraction-and-attention front end. The corpus is kept current: in addition to the historical 2015–2025 material we *refresh* it through the most recent twelve months by re-scraping both caucus archives—the Assembly Democratic Caucus (`asmdc.org`) and the Assembly Republican Caucus (`asmrc.org`)—from their published sitemaps,

the same uniform method for both parties (`fetch_refresh_releases.py`); the latest window analysed here runs to June 2026 (§2.7). The asymmetry between the two archives is itself informative and we report it plainly: the Democratic archive is comprehensive and date-stamped, while the Republican caucus site indexes far fewer releases and essentially no AI-focused ones in the window.

Index portability. Model-guided retrieval is not tied to one bibliographic index: the same claim-targeted queries run against *OpenAlex* or *Crossref*—the free DOI registry, every record a registered DOI by construction (`crossref_retrieval.py`)—and we report results under both, which establishes that the method depends on the model’s targeting rather than on any one service. Code, the benchmark harness, the committed judge labels and queries, the retrieved references, the scraped release text, and the figure scripts are public at <https://github.com/bdesmarais/ai-policy-science-use-public>, and the analysis regenerates from committed inputs with only the single frontier model.

Data and code availability

Code, the benchmark-validation harness, the committed judge labels and model-guided queries, the retrieved references, the scraped release text, and the figure scripts are publicly available at <https://github.com/bdesmarais/ai-policy-science-use-public>. The free components—OpenAlex and Crossref retrieval, the NLI surrogate, the open-source panel, the prediction-powered estimator, and all figures—regenerate from committed inputs with no paid service. The language steps depend on a single paid frontier model, Claude Opus 4.8 (claim extraction, retrieval guidance, and stance judging); its outputs are committed as static data and regenerate only with that model, exactly as frozen human labels would require the original coders. The validation benchmarks are SciFact, Climate-FEVER, and AVeriTeC.

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Author contributions

B.A.D. conceived and supervised the project. N.N. led the original data collection and pipeline development. B.A.D., S.R., J.J.H., and F.J.B. designed the study and the validation strategy. All authors interpreted the results. B.A.D. and N.N. drafted the manuscript, and all authors revised and approved it.

Competing interests

The authors declare no competing interests.

AI-assistance disclosure

This measurement instrument uses language models as components, and we disclose exactly which. A single frontier model, Claude Opus 4.8, performs the three language steps—claim extraction, retrieval guidance (targeted queries and abstention), and stance judging—used in context without

fine-tuning. Everything else is free and runs locally on a single consumer GPU: the scalable surrogate (DeBERTa- and BART-MNLI), the cross-developer panel (Qwen2.5, Phi-3.5, OLMo-2) and their combiners, the OpenAlex and Crossref indices, and the estimator. We also report how far the one paid component can itself be replaced: on the judging step a combined open stack reaches it without in-domain labels (§2.2). A high-capacity model also assisted in drafting and in executing the analysis; every cited reference was checked against a primary source, and all design decisions and claims are the authors’.

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